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1.2 Algebra & Functions

Laws of Indices

Indices (powers) follow a set of rules that allow us to simplify expressions involving powers.

Rule	Law	Example
Multiplication	$a^m \times a^n = a^{m+n}$	$x^3 \times x^4 = x^7$
Division	$a^m \div a^n = a^{m-n}$	$x^5 \div x^2 = x^3$
Power of a power	$(a^m)^n = a^{mn}$	$(x^2)^3 = x^6$
Zero index	$a^0 = 1$	$5^0 = 1$
Negative index	$a^{-n} = \frac{1}{a^n}$	$x^{-2} = \frac{1}{x^2}$
Fractional index	$a^{1/n} = \sqrt[n]{a}$	$8^{1/3} = 2$
General fractional	$a^{m/n} = (\sqrt[n]{a})^m$	$27^{2/3} = (\sqrt[3]{27})^2 = 9$

Example: Simplify $\frac{3x^{5/2} \times 2x^{-1}}{x^{1/2}}$

Numerator: $3x^{5/2} \times 2x^{-1} = 6x^{5/2-1} = 6x^{3/2}$

Divide by $x^{1/2}$: $6x^{3/2-1/2} = 6x$

Surds

A surd is an irrational root that cannot be simplified to a rational number, e.g. $\sqrt{2}, \sqrt{5}, \sqrt[3]{7}$.

Rule	Formula	Example
Multiplication	$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$	$\sqrt{3} \times \sqrt{12} = \sqrt{36} = 6$
Division	$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$	$\frac{\sqrt{20}}{\sqrt{5}} = \sqrt{4} = 2$
Simplification	$\sqrt{a^2 b} = a\sqrt{b}$	$\sqrt{50} = 5\sqrt{2}$
Rationalise (simple)	$\frac{1}{\sqrt{a}} = \frac{\sqrt{a}}{a}$	$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$
Rationalise (compound)	Multiply by conjugate $\frac{a \mp \sqrt{b}}{a \mp \sqrt{b}}$	$\frac{1}{3+\sqrt{2}} = \frac{3-\sqrt{2}}{7}$

Example: Rationalise $\frac{5}{2+\sqrt{3}}$

Multiply top and bottom by the conjugate $2 - \sqrt{3}$:

$$\frac{5}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} = \frac{5(2-\sqrt{3})}{4-3} = \frac{10-5\sqrt{3}}{1} = 10 - 5\sqrt{3}$$

Quadratics

A quadratic expression has the form $ax^2 + bx + c$ where $a \neq 0$. The graph is a parabola.

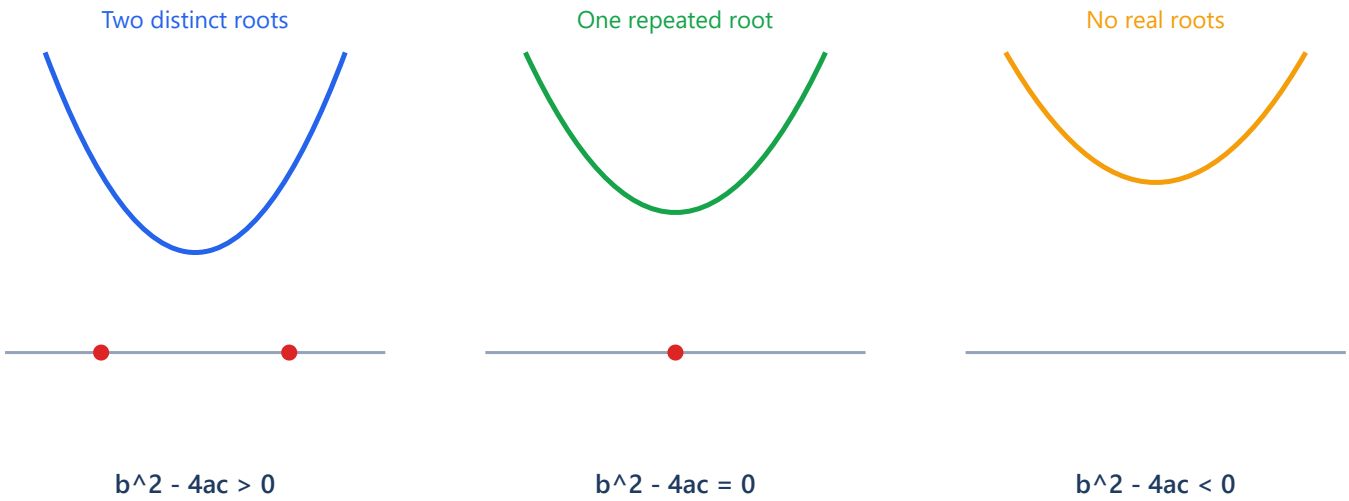
The Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The Discriminant

The **discriminant** $\Delta = b^2 - 4ac$ determines the nature of the roots of a quadratic equation.

Discriminant	Nature of Roots	Graph
$b^2 - 4ac > 0$	Two distinct real roots	Parabola crosses x-axis twice
$b^2 - 4ac = 0$	One repeated real root	Parabola touches x-axis once
$b^2 - 4ac < 0$	No real roots (complex)	Parabola does not cross x-axis



Completing the Square

$$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a} + c$$

Example: Write $2x^2 + 12x + 7$ in completed square form.

Factor out 2 from the x terms: $2(x^2 + 6x) + 7$

Complete the square inside: $2\big((x+3)^2 - 9\big) + 7$

Expand: $2(x+3)^2 - 18 + 7 = 2(x+3)^2 - 11$

Vertex is at $(-3, -11)$. Minimum value is -11.

Exam Tip: Completing the square tells you the vertex (turning point) of the parabola. If the question asks for the minimum/maximum value, this is the method to use. The vertex is $\left(-\frac{b}{2a},\ c - \frac{b^2}{4a}\right)$.

Simultaneous Equations

Linear-Linear

Use **elimination** (add/subtract equations) or **substitution**.

Linear-Quadratic

Substitute the linear equation into the quadratic and solve the resulting quadratic.

Example: Solve $y = x + 3$ and $x^2 + y^2 = 29$.

Substitute: $x^2 + (x+3)^2 = 29$

$x^2 + x^2 + 6x + 9 = 29$

$2x^2 + 6x - 20 = 0 \Rightarrow x^2 + 3x - 10 = 0$

$(x+5)(x-2) = 0 \Rightarrow x = -5$ or $x = 2$

When $x = -5$: $y = -2$. When $x = 2$: $y = 5$.

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Solutions: $(-5, -2)$ and $(2, 5)$.

Inequalities

Linear Inequalities

Solve like equations. **Reverse the inequality sign when multiplying/dividing by a negative number.**

Quadratic Inequalities

1. Solve the corresponding equation $ax^2 + bx + c = 0$ to find critical values.
2. Sketch the parabola.
3. Read off the solution from the graph.

Example: Solve $x^2 - 5x + 6 < 0$.

Factorise: $(x - 2)(x - 3) = 0$, so roots at $x = 2$ and $x = 3$.

The parabola is U-shaped (positive x^2 coefficient). It is below the x-axis between the roots.

Solution: $2 < x < 3$.

Exam Tip: Always sketch the quadratic to decide the correct region. For > 0 with a U-shaped parabola, the answer is two separate regions. For < 0, the answer is between the roots.

Polynomials

Factor Theorem

If $f(a) = 0$, then $(x - a)$ is a factor of $f(x)$.
Conversely, if $(x - a)$ is a factor of $f(x)$, then $f(a) = 0$.

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Example: Show that $(x-2)$ is a factor of $f(x) = x^3 - 6x^2 + 11x - 6$.

$f(2) = 8 - 24 + 22 - 6 = 0$. Since $f(2) = 0$, $(x-2)$ is a factor. Then:

$x^3 - 6x^2 + 11x - 6 = (x-2)(x^2 - 4x + 3) = (x-1)(x-2)(x-3)$

Algebraic Division

Use polynomial long division or compare coefficients to divide a polynomial by a linear or quadratic factor.

Example: Divide $2x^3 + 3x^2 - 8x + 3$ by $(x+3)$.

Using long division or inspection:

$$2x^3 + 3x^2 - 8x + 3 = (x+3)(2x^2 - 3x + 1) = (x+3)(2x-1)(x-1)$$

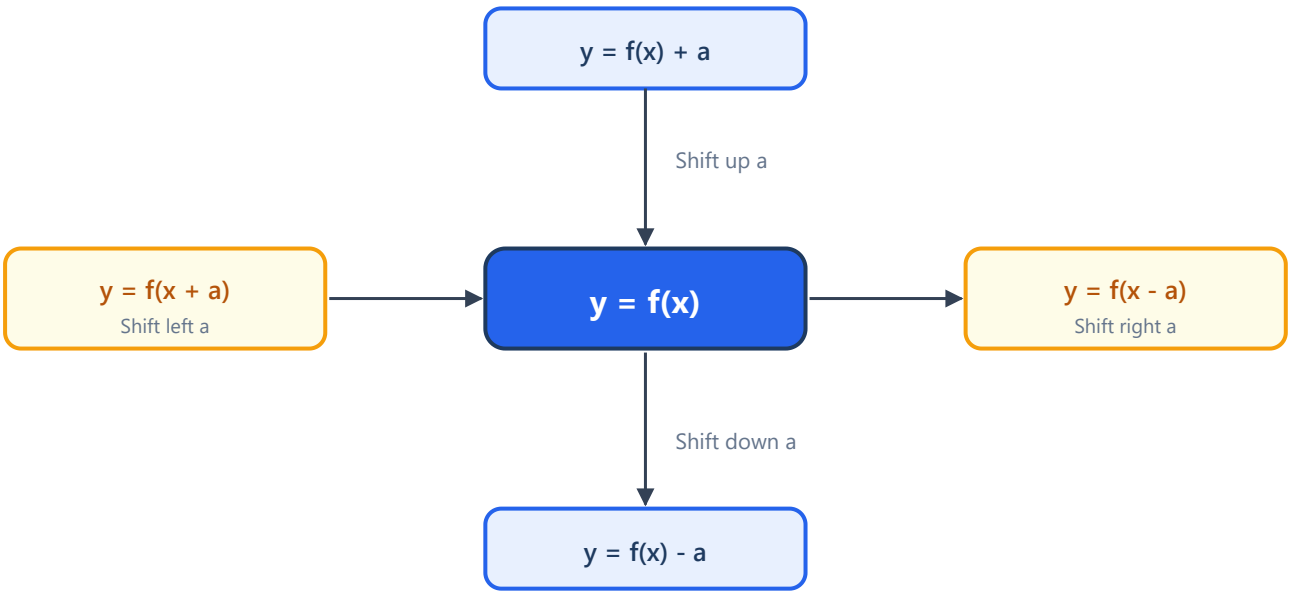
Graphs of Functions

You need to recognise and sketch the following key graphs:

- Linear: $y = mx + c$
- Quadratic: $y = ax^2 + bx + c$
- Cubic: $y = ax^3 + \dots$
- Reciprocal: $y = \dfrac{a}{x}$ and $y = \dfrac{a}{x^2}$
- Exponential: $y = a^x$
- Trigonometric: $y = \sin x, \cos x, \tan x$

Graph Transformations

Transformation	Notation	Effect
Translation up by a	$y = f(x) + a$	Shift graph up a units
Translation down by a	$y = f(x) - a$	Shift graph down a units
Translation right by a	$y = f(x - a)$	Shift graph right a units
Translation left by a	$y = f(x + a)$	Shift graph left a units
Vertical stretch, scale factor a	$y = af(x)$	Stretch parallel to y-axis by factor a
Horizontal stretch, scale factor $\frac{1}{a}$	$y = f(ax)$	Stretch parallel to x-axis by factor $\frac{1}{a}$
Reflection in x-axis	$y = -f(x)$	Flip graph over x-axis
Reflection in y-axis	$y = f(-x)$	Flip graph over y-axis



Exam Tip: Remember the counter-intuitive rule: changes *inside* the bracket (to x) do the *opposite* of what you expect. $f(x - 3)$ shifts **right** by 3, and $f(2x)$ squashes the graph **horizontally** by factor $\frac{1}{2}$.

Partial Fractions

Partial fractions decompose a single fraction into a sum of simpler fractions. This is the reverse of adding fractions together.

Types

Denominator Type	Partial Fraction Form
Linear factors: $\dfrac{f(x)}{(ax+b)(cx+d)}$	$\dfrac{A}{ax+b}+\dfrac{B}{cx+d}$
Repeated factor: $\dfrac{f(x)}{(ax+b)^2(cx+d)}$	$\dfrac{A}{ax+b}+\dfrac{B}{(ax+b)^2}+\dfrac{C}{cx+d}$
Improper (degree num $\geq$ denom)	Divide first, then decompose the remainder

Example: Express $\dfrac{5x+1}{(x-1)(x+2)}$ in partial fractions.

Let $\dfrac{5x+1}{(x-1)(x+2)}=\dfrac{A}{x-1}+\dfrac{B}{x+2}$

Multiply through: $5x+1=A(x+2)+B(x-1)$

Set $x=1$: $6=3A\Rightarrow A=2$

Set $x=-2$: $-9=-3B\Rightarrow B=3$

$$\frac{5x+1}{(x-1)(x+2)}=\frac{2}{x-1}+\frac{3}{x+2}$$

Modulus Function

The **modulus** (absolute value) of x, written $|x|$, gives the magnitude of x without regard to sign.

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

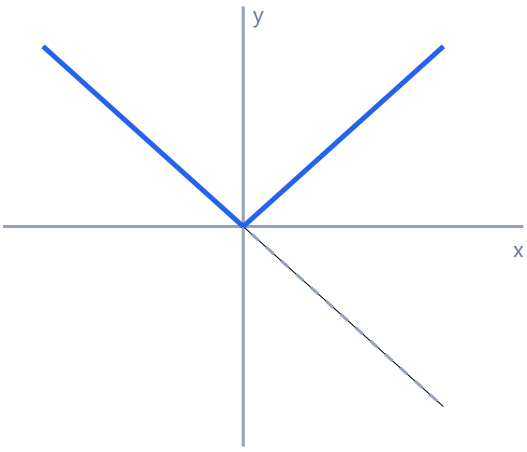
Graph of $y = |f(x)|$

Reflect any part of $y = f(x)$ that is below the x-axis in the x-axis (make negative y-values positive).

Graph of $y = f(|x|)$

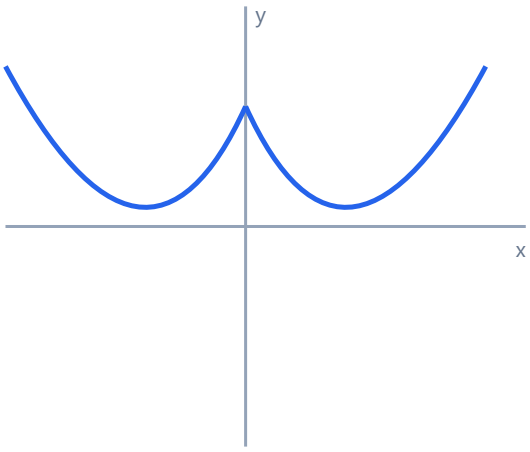
Keep the graph for $x \geq 0$ and reflect it in the y-axis (the graph becomes symmetric about the y-axis).

y = |f(x)|



Reflect below x-axis upward

y = f(|x|)



Symmetric about y-axis

Solving Modulus Equations

Example: Solve $|2x - 3| = 5$.

Case 1: $2x - 3 = 5 \Rightarrow 2x = 8 \Rightarrow x = 4$

Case 2: $2x - 3 = -5 \Rightarrow 2x = -2 \Rightarrow x = -1$

Solutions: $x = 4$ or $x = -1$.

Example: Solve $|x+1| = |2x-3|$.

Case 1: $x + 1 = 2x - 3 \Rightarrow 4 = x \Rightarrow x = 4$

Case 2: $x + 1 = -(2x - 3) \Rightarrow x + 1 = -2x + 3 \Rightarrow 3x = 2 \Rightarrow x = \frac{2}{3}$

Solutions: $x = 4$ or $x = \frac{2}{3}$.

Exam Tip: When solving $|f(x)| = |g(x)|$, you can also square both sides: $f(x)^2 = g(x)^2$, which avoids case work. For modulus inequalities like $|f(x)| < a$, rewrite as $-a < f(x) < a$.

Functions: Domain and Range

- The **domain** of a function is the set of all valid input values (x-values).
- The **range** (codomain) is the set of all possible output values ($f(x)$-values).
- A function maps each element of the domain to **exactly one** element of the range.

Function	Natural domain	Range
$f(x) = x^2$	$x \in \mathbb{R}$	$f(x) \geq 0$
$f(x) = \sqrt{x}$	$x \geq 0$	$f(x) \geq 0$
$f(x) = \dfrac{1}{x}$	$x \in \mathbb{R},\ x \neq 0$	$f(x) \in \mathbb{R},\ f(x) \neq 0$
$f(x) = \ln x$	$x > 0$	$f(x) \in \mathbb{R}$
$f(x) = \sqrt{9 - x^2}$	$-3 \leq x \leq 3$	$0 \leq f(x) \leq 3$

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Exam Tip: Common restrictions on the domain:

- Cannot divide by zero: exclude values making the denominator zero.
- Cannot take the square root of a negative number (over the reals): set the expression ≥ 0.
- Cannot take $\ln$ of a non-positive number: set the argument > 0.

Composite Functions

The **composite function** $fg(x)$ (read "f of g of x") means apply g first, then apply f to the result: \[fg(x) = f(g(x))\] In general, $fg(x) \neq gf(x)$ — order matters.

Example: Given $f(x) = 2x + 1$ and $g(x) = x^2$, find $fg(x)$ and $gf(x)$.

$fg(x) = f(g(x)) = f(x^2) = 2x^2 + 1$

$gf(x) = g(f(x)) = g(2x + 1) = (2x + 1)^2 = 4x^2 + 4x + 1$

These are different functions.

Example: $f(x) = \sqrt{x - 1}$ (domain $x \geq 1$) and $g(x) = x^2 + 1$. Find $fg(x)$ and its domain.

$fg(x) = f(g(x)) = f(x^2 + 1) = \sqrt{(x^2 + 1) - 1} = \sqrt{x^2} = |x|$

The domain of fg requires $g(x) \geq 1$, i.e. $x^2 + 1 \geq 1$, which is true for all $x \in \mathbb{R}$.

So $fg(x) = |x|$, domain $x \in \mathbb{R}$, range $fg(x) \geq 0$.

Exam Tip: The domain of fg is restricted by two conditions: (1) x must be in the domain of g, and (2) $g(x)$ must be in the domain of f. Apply both conditions and take the intersection.

Inverse Functions

The **inverse function** $f^{-1}(x)$ reverses the action of f. It exists only when f is a **one-to-one** (injective) function — every output value corresponds to exactly one input value.

Key properties:

- $f^{-1}(f(x)) = x$ and $f(f^{-1}(x)) = x$
- The domain of f^{-1} is the range of f, and vice versa.
- The graph of $y = f^{-1}(x)$ is the reflection of $y = f(x)$ in the line $y = x$.

Finding an Inverse Function

1. Write $y = f(x)$.
2. Rearrange to make x the subject.
3. Replace x with $f^{-1}(x)$ and y with x.
4. State the domain of f^{-1} (= range of f).

Example: Find $f^{-1}(x)$ for $f(x) = 3x - 2$.

Let $y = 3x - 2$. Rearrange: $x = \dfrac{y+2}{3}$.

So $f^{-1}(x) = \dfrac{x+2}{3}$, domain $x \in \mathbb{R}$.

Check: $f(f^{-1}(x)) = 3 \cdot \dfrac{x+2}{3} - 2 = x + 2 - 2 = x$ ✓

Example: $f(x) = \dfrac{2x+1}{x-3}$ $(x \neq 3)$. Find $f^{-1}(x)$.

Let $y = \dfrac{2x+1}{x-3}$. Cross multiply: $y(x-3) = 2x + 1$.

$xy - 3y = 2x + 1 \Rightarrow xy - 2x = 3y + 1 \Rightarrow x(y-2) = 3y + 1$.

$$x = \frac{3y+1}{y-2}$$

So $f^{-1}(x) = \dfrac{3x+1}{x-2}$, domain $x \neq 2$.

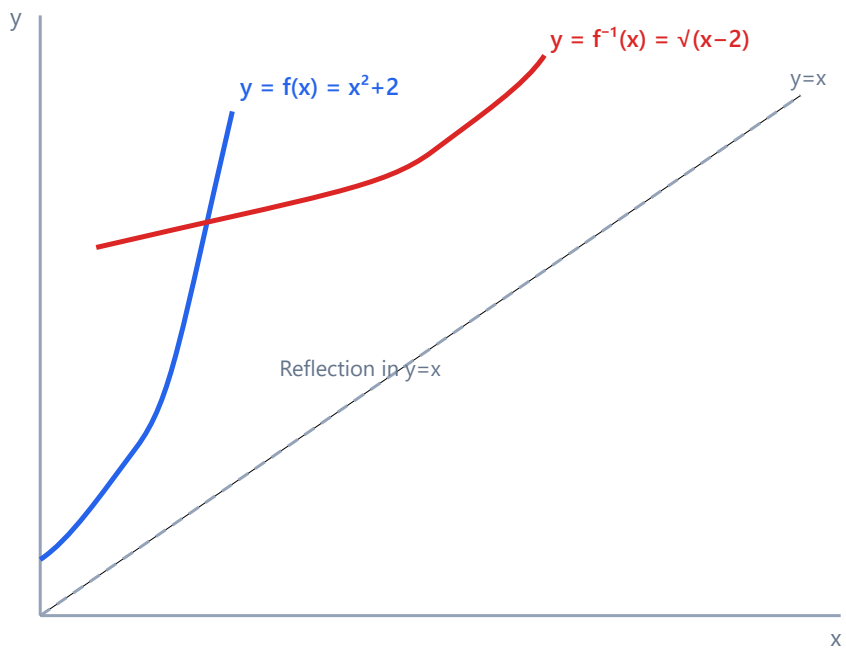
Note: f^{-1} has the same algebraic form structure as f (a common feature of this type of function).

Example: $f(x) = x^2 + 2$, domain $x \geq 0$. Find $f^{-1}(x)$ and its domain.

The domain restriction $x \geq 0$ makes f one-to-one (without it, f would not have an inverse).

Let $y = x^2 + 2$. Then $x = \sqrt{y-2}$ (taking positive root since $x \geq 0$).

$f^{-1}(x) = \sqrt{x-2}$, domain: $x \geq 2$ (= range of f).



Exam Tip: A function only has an inverse if it is **one-to-one**. For many-to-one functions (like x^2 on all reals), you must restrict the domain first. The examiner may give you a domain — use it. Always state the domain of the inverse function.

Self-Inverse Functions

A function is **self-inverse** if $f^{-1}(x) = f(x)$, i.e. $f(f(x)) = x$. Examples include $f(x) = \frac{1}{x}$ and $f(x) = \frac{a}{x}$. Their graphs are symmetric about $y = x$.